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Economic Policy Uncertainty and China's FDI Inflows: Moderating Effects of Financial Development and Political Stability

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Abstract

This paper investigates the impact of global EPU and China's EPU on China's FDI inflows, examining whether financial development and political stability moderate these relationships. Using panel data from 212 countries spanning 2009 to 2022, we first establish causal direction through Granger causality tests, then employ instrumental variable estimation to address endogeneity concerns, while conducting heterogeneity analysis across development levels and Belt and Road Initiative participation. We find that both global and domestic EPU significantly reduce China's FDI inflows, with a 1% increase in China's EPU leading to a 0.083% decrease in FDI inflows. However, political stability and financial development serve as effective moderators, reducing EPU's negative impact by up to 60% and 70%, respectively. The effects vary substantially across investor countries: non-developed countries show ten times stronger sensitivity to EPU than developed countries, while Belt and Road Initiative countries demonstrate 86% lower sensitivity than non-BRI countries. This research advances EPU–FDI theory by demonstrating how institutional quality creates "policy buffers" against uncertainty and provides policymakers with evidence that strengthening political stability and financial development can maintain investor confidence during uncertain periods, while strategic international partnerships can insulate investment flows from policy volatility.

Keywords: foreign direct investment; economic policy uncertainty; financial development; political stability

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1. Introduction

Foreign direct investment (FDI) has been one of the main drivers of economic growth and plays a crucial role in both host and home countries. However, following the global financial crisis in 2008, global FDI growth began to stagnate and fell sharply in 2020 to below USD 1 trillion, representing a decline of 34.7% due to the pandemic. China's FDI has been significantly affected as a result. By the end of September 2023, foreign firms had withdrawn a total of more than USD 160 billion from China for six consecutive quarters. This has caused the amount of FDI in China to fall into negative territory for the first time in approximately 25 years.

Following the global financial crisis, we have observed that economic policy uncertainty is a significant factor influencing FDI inflows. Governments have frequently adjusted policies to stimulate the economy, leading to an unstable policy environment and heightened macroeconomic volatility (Colombo, 2013; Karnizova & Li, 2014). Currently, there is

limited understanding of the impact of policy environment volatility on economic activity compared to other economic factors. Previous studies have extensively explored the influence of various factors on FDI inflows, primarily focusing on long-term structural determinants. Researchers have consistently examined GDP (L. K. Cheng & Kwan, 2000), infrastructure development (Shah, 2014), and trade openness (Demirhan & Masca, 2008) as the primary drivers of FDI flows. However, these traditional determinants represent relatively stable, long-term characteristics that change gradually over several years, whereas EPU, as a more dynamic, short-term factor, may fluctuate significantly within months or quarters.

Furthermore, the existing literature presents inconsistent conclusions regarding the positive or negative impact of policy-related economic uncertainty on FDI inflows. First, from the perspective of host country EPU, some scholars argue that host country EPU has a negative impact on FDI inflows (Chen & Funke, 2011; Pennings & Sleuwaegen, 2004). Others disagree, with researchers supporting the growth choice theory arguing that increased host country economic policy uncertainty has a positive impact on FDI inflows (Hartman, 1972; Oi, 1961; Vo & Le, 2017). Knight (1921) argued that firms can identify and capitalize on investment opportunities to generate profits by integrating resources in an uncertain economic policy environment. Investors tend to focus on potential opportunities and returns and increase their investment efforts when uncertainty rises.

Furthermore, while previous studies have extensively examined the impact of economic policy adjustments (EPU) on macroeconomic variables and investment behavior, few have focused on the role of global EPU in influencing FDI inflows. Existing research primarily centers on single-dimensional analyses of domestic EPU, such as Gao et al. (2024), who specifically studied the impact of domestic EPU in 264 Chinese cities on FDI inflows; C. H. J. Cheng (2017) analyzed the role of domestic policy uncertainty shocks on economic fluctuations in South Korea; and Sinha and Ghosh (2021) studied the impact of domestic EPU on FDI inflows in India. Regarding international uncertainty, some scholars have pointed out that the impact of global EPU on local financial markets is a rarely explored angle (Gong et al., 2023). Some studies have found that, in addition to domestic uncertainty, uncertainty in other regions/countries can also affect a country's economic activity (e.g., Colombo (2013), Carrière-Swallow and Céspedes (2013), C. H. J. Cheng (2017)). These findings suggest that international investors consider not only the host country's EPU but also the EPU of other regions when making FDI investment decisions (Canh et al., 2020).

Although the importance of institutional factors is increasingly recognized, Chinh and Thi Minh Hue (2025) found that while 'improvements in institutional quality and financial development have a positive impact on FDI, the interaction between these two factors may produce opposite effects,' indicating that the moderating mechanism of the joint effects of political stability and financial development remains poorly understood (Bommadevara & Sakharkar, 2021).

Therefore, from the perspective of the host country's EPU, related studies have not obtained consistent conclusions. We need to conduct a further systematic and comprehensive analysis to explore the impact of EPU on China's FDI inflows. In this paper, taking China as an example, the specific research questions are as follows: (1) Whether and how world EPU affects China's FDI? (2) Whether and how China's EPU affects China's FDI inflows? (3) Whether and how financial development and political stability moderate the relationship between China's EPU and China's FDI inflows? The specific objectives of this study are as follows: (1) To determine the impact of world EPU on China's FDI inflows. (2) To determine the impact of China's EPU on China's FDI inflows. (3) To determine the moderating role of financial development and political stability on the relationship between China's EPU and China's FDI inflows.

This study contributes to the theory and empirical evidence in the following aspects. First, it provides insight for solving the problem of China's continuous withdrawal of foreign capital and explaining the phenomenon of China's declining FDI. It enriches the application of real options theory, precautionary savings theory, and financial friction theory in the Chinese context. Second, against the backdrop of high global and China's EPU, the exploration of the moderating effects of financial development and political stability can provide insights into mitigating this phenomenon. Third, this study can provide some implications for national policymakers.

The rest of the paper proceeds as follows: Section 2 presents the findings of the relevant literature; Section 3 describes the data sources and the empirical model; Section 4 includes the empirical results and robustness tests; the final section presents the conclusions and implications.

2. Literature Review

2.1. Global Economic Policy Uncertainty and Foreign Direct Investment Inflows

The relationship between global economic policy uncertainty (GEPU) and FDI inflows has generated considerable academic interest, though findings remain mixed. Several studies have found a consistently negative effect of global EPU on FDI, particularly in emerging economies and regions such as sub-Saharan Africa. For example, Avom et al. (2020) emphasize how increased global uncertainty significantly deters foreign investment flows in less developed contexts. Colombo (2013) further reinforces this view, arguing that GEPU tends to exert a stronger deterrent effect on FDI than domestic uncertainty in many cases.

Conversely, other scholars have offered more nuanced interpretations. Canh et al. (2020), for instance, argue that while domestic EPU tends to discourage FDI, higher levels of global EPU may, in some cases, redirect FDI toward relatively more stable countries. This counterintuitive behavior is often explained through behavioral economics—specifically, the "anchoring and adjustment" bias, whereby investors overemphasize familiar benchmarks when assessing uncertainty.

In the context of China, persistently elevated global EPU following the pandemic may have contributed to a more cautious investment climate, limiting the prospects for capital inflows. Such developments point to the importance of distinguishing between domestic and international sources of policy uncertainty when evaluating their impact on FDI. Based on this, we propose the following hypotheses:

H1. *Global EPU has a significant causal effect on FDI inflows to China.*

H2. *An increase in global EPU has a significant negative effect on China's FDI inflows.*

2.2. China's Domestic Economic Policy Uncertainty and Foreign Direct Investment Inflows

While all forms of investment involve uncertainty, foreign direct investment is particularly sensitive to policy instability due to its irreversible nature and long-term orientation. Compared to domestic investment, FDI decisions typically involve higher sunk costs, longer planning horizons, and greater exposure to legal and regulatory risks (Choi et al., 2021; Julio & Yook, 2016). Additional challenges such as exchange rate volatility, capital repatriation restrictions, and limited access to local legal protection further heighten risk perceptions for foreign investors (Aizenman & Spiegel, 2006; Gulen & Ion, 2015).

In such a context, elevated domestic EPU may deter investment by amplifying the so-called "liability of foreignness"—a term used to describe the disadvantages foreign firms face compared to domestic competitors (Zaheer & Mosakowski, 1997). Conse-

quently, investors may choose to delay or forgo investment until a more predictable policy environment emerges.

That said, not all scholars agree that policy uncertainty necessarily reduces FDI. Some researchers, drawing on real options and growth option theories, argue that in certain cases, heightened uncertainty might attract investors seeking high-risk, high-reward opportunities (Hartman, 1972; Oi, 1961; Vo & Le, 2017). Knight (1921) posited that firms willing to take calculated risks during volatile periods might be better positioned to consolidate resources and capture long-term profits.

Despite these contrasting views, the prevailing empirical evidence tends to support a negative link between domestic EPU and FDI. For instance, Shahzad and Al-Swidi (2013) observed that lower levels of policy uncertainty enhanced investor confidence in Pakistan, while Avom et al. (2020) found that EPU exerted a more severe effect on FDI in emerging economies than in developed ones. Similarly, Zhu et al. (2019) demonstrated that economic uncertainty significantly reduced FDI in their panel of 23 countries. Accordingly, the following hypotheses are proposed:

H3. *China's domestic EPU has a significant causal effect on its FDI inflows.*

H4. *An increase in China's domestic EPU has a significant negative effect on FDI inflows.*

2.3. Moderating Factors in EPU–FDI Relationship

Moderating factors, such as political stability and financial development, can influence the EPU–FDI dynamic:

Strong political stability attracts FDI by providing stability and reducing risks (Busse & Hefeker, 2007; Feng & Mu, 2010; Warner & Zawahri, 2012; Wu & Shao, 2023). Conversely, less mature environments may still draw investment through relaxed trade policies (Ahlquist & Prakash, 2010; Shirodkar & Mohr, 2015).

Financial development provides a 'cushion' for foreign-owned firms by improving the efficiency of capital allocation and reducing financing constraints, enabling them to better cope with shocks caused by policy uncertainty (Yu et al., 2024). A highly developed financial system also facilitates the flow of information, helping investors to more accurately assess the material impact of policy changes rather than overreacting.

Accordingly, the following hypotheses are proposed:

H5. *Higher levels of political stability have a significant positive moderating effect on the relationship between EPU and FDI inflows.*

H6. *Higher levels of financial development have a significant positive moderating effect on the relationship between EPU and FDI inflows.*

3. Data and Methodology

3.1. Data Sources and Variable Definitions

To examine how both global and domestic economic policy uncertainty (EPU) impact China's foreign direct investment (FDI) inflows, we compiled a comprehensive panel dataset covering 212 countries from 2009 to 2022, amounting to 2968 observations.

FDI inflow data to China were obtained from the International Monetary Fund (IMF), while EPU indices—both global and China-specific—were sourced from the EPU database developed by Baker et al. (2016). Supplementary macroeconomic indicators were retrieved from the World Bank's World Development Indicators (WDI) and the China Statistical Yearbook.

Table 1 presents the key variables used in the study, along with their definitions and sources. The dependent variable is the logarithmic form of FDI inflows to China. The key independent variables are the global EPU index (GEPU) and China's domestic EPU index (CEPU). Control variables include trade openness, economic growth, market size, infrastructure, human capital, inflation, resource endowment, high-tech export ratio, carbon emissions, and institutional indicators such as political stability (ICRG) and financial development.

Table 1. Variable Description.

Variable	Variable Meaning	Variable Description	Data Source
FDI	FDI inflows to China	China's FDI flows	IMF
EPU	China's or other countries' economic policy uncertainty	Economic policy uncertainty index	EPU website
Market	Economies of scale	To measure China's market size as the host country for FDI, we use China's GDP.	World Bank
Open	The degree of trade openness	To measure the openness, we use the proportion of total trade in GDP.	World Bank
Growth	China's level of economic development	To measure China's level of economic development, we use China's per capita GDP.	World Bank
EduP	The level of human capital	Percentage of employed persons who are university students and above	Statistical Yearbook
Infra	Status of infrastructure	To measure the state of a country's infrastructure, we utilize the logarithmic form of highway freight volume.	Statistical Yearbook
Re	The natural resource endowment of the host country	The share of mineral and metal resources exported by host countries in total commodity exports	World Bank
Tec	The strategic resource endowment of the host country	The proportion of exports of high-tech products in the export value of manufactured goods	World Bank
Inflation	Inflation rates in the host country	Annual inflation rate as measured by the consumer price index	World Bank
P_CO2	Carbon dioxide emissions per capita	Carbon dioxide emissions per capita	World Bank
ICRG	Degree of political stability in China	Political Stability Index (PSI)	PRS Database
Financial	The extent to which banks are able to provide financial resources to investors	Financial development is defined as domestic credit to the private sector as a share of GDP	IMF

This broad set of variables allows us to control for country-specific macroeconomic and structural differences that might influence FDI inflows.

3.2. Descriptive Statistics and Correlations

Table 2 presents the descriptive statistics of the key variables in this paper. The average FDI is 3.259, with a standard deviation of 3.668, indicating significant variations in FDI across different regions. The China Economic Policy Uncertainty index (China_EPU) exhibits a mean of 5.142 with a standard deviation of only 0.511, suggesting relative stability of this indicator. The Market index shows a mean of 13.464, approaching its theoretical maximum and indicating high overall marketization levels across sample regions. Among control variables, technology level (Tec) displays the greatest variability (standard deviation of 1.102), reflecting substantial regional differences in technological development, while infrastructure level (Infra) shows smaller variation, indicating more balanced development across regions. The consistent number of observations across all variables ensures data completeness.

Table 2. Summary Statistics.

Variable	Obs	Mean	SD	Min	Median	Max
FDI	2968	3.259	3.668	−8.356	2.524	14.493
China_EPU	2968	5.142	0.511	4.523	4.848	5.967
Market	2968	13.464	0.367	12.760	13.478	13.996
Open	2968	0.386	0.062	0.320	0.354	0.492
Growth	2968	10.845	0.349	10.173	10.855	11.359
EduP	2968	17.118	4.846	7.430	17.750	24.056
Infra	2968	14.979	0.168	14.571	14.998	15.191
Re	2968	1.554	0.224	1.227	1.547	1.885
Tec	2968	50.182	1.102	48.013	50.177	52.155
ICRG	2968	8.815	0.732	7.542	8.729	10.500
Inlation	2968	2.189	1.295	−0.703	2.037	5.411
P_CO2	2968	8.592	1.173	6.221	8.788	11.186
Financial	2968	5.005	0.140	4.810	5.035	5.222

Notes: This table presents descriptive statistics for key variables in the econometric model, including the number of observations (Obs), mean, standard deviation (SD), minimum (Min), median (Med), and maximum (Max).

Table 3 presents the Pearson correlation matrix. Notably, both global and China-specific EPU indices are negatively correlated with FDI inflows. This preliminary evidence supports the study's hypothesis that uncertainty deters foreign investment. High correlations among some control variables—such as GDP and infrastructure—highlight the need for robust regression techniques to avoid multicollinearity.

Table 3. Correlation Matrix.

	FDI	China_EPU	GEPU	lnGDP	Open	Growth	EduP	Infra	Re	Tec	Inflation	P_CO2
FDI	1											
China_EPU	−0.019	1										
GEPU	−0.017	0.937 ***	1									
lnGDP	−0.011	0.789 ***	0.820 ***	1								
Open	0.018	−0.684 ***	−0.677 ***	−0.845 ***	1							
Growth	−0.011	0.789 ***	0.822 ***	1.000 ***	−0.838 ***	1						
EduP	−0.015	0.776 ***	0.833 ***	0.990 ***	−0.827 ***	0.990 ***	1					
Infra	0.002	0.627 ***	0.666 ***	0.919 ***	−0.736 ***	0.918 ***	0.886 ***	1				
Re	−0.003	0.229 ***	0.059 ***	−0.109 ***	0.199 ***	−0.108 ***	−0.131 ***	−0.119 ***	1			
Tec	−0.002	0.349 ***	0.184 ***	−0.207 ***	0.103 ***	−0.207 ***	−0.247 ***	−0.313 ***	0.466 ***	1		
Inflation	0.018	−0.045 **	−0.057 ***	−0.181 ***	0.481 ***	−0.173 ***	−0.196 ***	−0.036 **	0.304 ***	0.334 ***	1	
P_CO2	−0.005	0.528 ***	0.639 ***	0.887 ***	−0.659 ***	0.890 ***	0.889 ***	0.834 ***	−0.137 ***	−0.518 ***	−0.276 ***	1

Pearson's correlation coefficients; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

3.3. The Empirical Strategy

3.3.1. Model Selection and Justification

Panel Data Model Rationale

This study employs panel regression models for several compelling theoretical and empirical reasons. First, the research question fundamentally requires exploiting both cross-sectional variation (differences across source countries) and temporal variation (changes over time in EPU and FDI flows) to identify how uncertainty affects investment decisions. Panel data allow us to capture the heterogeneous responses of different countries to China's

policy uncertainty while controlling for time-invariant country characteristics that might confound our estimates (Baltagi, 2021; Wooldridge, 2002).

Second, FDI decisions are inherently dynamic processes influenced by both observable factors (GDP, trade openness, infrastructure) and unobservable characteristics (cultural ties, historical relationships, institutional trust) (Blonigen & Piger, 2014). A simple cross-sectional analysis would suffer from omitted variable bias, while pure time-series analysis would ignore the rich heterogeneity across source countries. Panel data methodology enables us to control for unobserved heterogeneity through country-specific effects while maintaining sufficient variation for identification (Hsiao, 2018).

Third, the panel structure allows us to address potential reverse causality concerns. While our primary interest lies in how EPU affects FDI, large FDI flows might theoretically influence policy uncertainty through political economy channels (Carkovic & Levine, 2005). The panel framework enables us to implement instrumental variable strategies and examine temporal precedence more rigorously than alternative approaches (Cameron & Trivedi, 2005).

Model Assumptions and Validity

The panel regression approach relies on several key assumptions. The strict exogeneity assumption requires that error terms are uncorrelated with explanatory variables across all time periods (Stock & Watson, 2018). Our research addresses this through our instrumental variable strategy discussed below. The homoskedasticity assumption is relaxed through robust standard error estimation clustered at the country level to account for potential correlation in errors within countries over time (Cameron et al., 2011; Marti et al., 2015).

Most critically, we assume that unobserved heterogeneity follows a specific structure. Country-fixed effects capture time-invariant characteristics such as geographical distance, cultural affinity, and historical ties with China (Mundlak, 1978). Time-fixed effects control for global shocks affecting all countries simultaneously, such as the 2008 financial crisis or COVID-19 pandemic (Pesaran, 2006). The remaining idiosyncratic error is assumed to be independently distributed, which we verify through diagnostic tests (Wooldridge, 2002).

Identification of Endogenous Problems

The relationship between EPU and FDI faces several potential endogeneity challenges that we address comprehensively (Antonakakis et al., 2017; Canh et al., 2020):

Reverse Causality: Large FDI outflows from specific countries to China might influence China's domestic policy uncertainty through lobbying, political pressure, or economic interdependence effects (Gulen & Ion, 2015). While this channel seems empirically limited given the relative scale of individual country FDI flows compared to China's economy, we cannot a priori dismiss this possibility (Julio & Yook, 2016).

Omitted Variable Bias: Unobserved factors affecting both EPU and FDI could bias our estimates. For example, global risk sentiment might simultaneously increase EPU indices and reduce international investment flows (Bloom, 2009; Carrière-Swallow & Céspedes, 2013). Our time-fixed effects partially address global confounders, but country-specific time-varying omitted variables remain a concern.

Measurement Error: The EPU index, constructed from newspaper coverage, might imperfectly capture true policy uncertainty, potentially leading to attenuation bias in our estimates (Azzimonti, 2018; Baker et al., 2016).

3.3.2. Instrumental Variable Strategy

Instrumental Variable Selection

We employ lagged values of EPU as instrumental variables, specifically EPU(t−1) as an instrument for EPU(t). This choice is motivated by several theoretical and empirical considerations (J. D. Angrist & Pischke, 2009; Murray, 2006):

Relevance Condition: Economic policy uncertainty exhibits significant persistence, with current uncertainty levels strongly predicted by recent past values (Baker et al., 2016; Bloom, 2009). This persistence reflects the gradual evolution of policy environments and the sticky nature of institutional changes (Acemoglu & Robinson, 2013). The first-stage F-statistics consistently exceed conventional thresholds ($F > 10$), confirming instrument relevance (Hausman et al., 2005).

Exclusion Restriction: The key assumption is that lagged EPU affects current FDI only through its impact on current EPU, not through direct channels. This assumption is plausible because: (1) FDI decisions typically respond to current and anticipated future uncertainty rather than purely historical uncertainty (Dixit & Pindyck, 1994); (2) the one-year lag ensures that any direct effects of past uncertainty on current investment decisions have been absorbed through current uncertainty measures; (3) investors' memories and decision-making processes focus primarily on recent and current information rather than increasingly distant past signals (Kahneman & Tversky, 1984).

Instrumental Variable Validity and Limitations

While lagged EPU represents the most feasible instrumental variable approach in our context, we acknowledge several limitations (Bound et al., 1995; Staiger & Stock, 1997):

Potential Violations: If FDI decisions exhibit longer memory effects, lagged EPU might violate the exclusion restriction through direct channels (Conley et al., 2012). However, given the rapid pace of international capital markets and the forward-looking nature of investment decisions, this concern appears limited (Blonigen & Piger, 2014).

Weak Instrument Concerns: Although our F-statistics exceed conventional thresholds, the persistence of EPU means our instrument might not provide fully exogenous variation (Andrews et al., 2019). We address this through robust inference procedures and by examining the sensitivity of results to alternative lag structures (Olea & Pflueger, 2013).

Limited External Validity: The instrumental variable approach identifies local average treatment effects for countries whose FDI decisions are responsive to EPU persistence patterns (J. Angrist & Imbens, 1995). This effect might not generalize to all countries or time periods.

Alternative Identification Strategies Considered

We considered several alternative approaches before settling on our instrumental variable strategy (J. D. Angrist & Pischke, 2009):

Natural Experiments: Major policy announcements or leadership changes could provide exogenous variation in EPU (Dube et al., 2011). However, such events are rare, country-specific, and might affect FDI through channels beyond uncertainty, limiting their validity as instruments.

External Instruments: Using EPU from other countries as instruments for China's EPU was considered but rejected due to high correlation (suggesting common global factors) and questionable exclusion restrictions (global uncertainty might directly affect FDI to China) (Canh et al., 2020).

Dynamic Panel Methods: System GMM approaches could address endogeneity through internal instruments but require strong assumptions about the error structure

and perform poorly with persistent dependent variables like FDI flows (Bond et al., 2001; Roodman, 2009).

Regulatory Policy Shocks: Specific policy changes affecting uncertainty could serve as instruments, but identifying truly exogenous policy shocks remains challenging, and such approaches might capture direct policy effects rather than pure uncertainty channels (Hassan et al., 2019).

Given these alternatives' limitations, lagged EPU emerges as the most credible identification strategy, providing a reasonable balance between theoretical validity and empirical feasibility.

3.3.3. Granger Causality Test Model

To test for directional causality between EPU and FDI inflows, we apply the Granger causality framework using a Vector Autoregressive (VAR) model. The approach involves two stages:

First, we regress FDI inflows on their own lagged values (restricted model).

Next, we add lagged values of EPU (either GEPU or CEPU) to create an unrestricted model.

The F-statistic from comparing these two regressions determines whether EPU significantly improves the model's predictive power. A significant result implies that EPU Granger-causes changes in FDI inflows. We repeat this process for both global and domestic EPU indicators.

3.3.4. Baseline Panel Regression Model

To quantify the relationship between economic policy uncertainty and FDI inflows to China, we estimate the following baseline regression models:

$$\ln(FDI_{it}) = \alpha + \beta_1 \ln(GEPU_t) + \beta_2 X_{ct} + \varepsilon_{ct} \quad (1)$$

$$\ln(FDI_{it}) = \alpha + \beta_1 \ln(CEPU_t) + \beta_2 X_{ct} + \varepsilon_{ct} \quad (2)$$

where subscripts i, c, and t stand for home country, China, and year, respectively. $\ln(FDI_{it})$ denotes the logarithmic form of FDI to China for different country i in year t. The dependent variable $\ln(GEPU_t)$ and $ln(CEPU_t)$, which measures the logarithmic form of economic policy uncertainty for the world and China separately in year t. X_{ct} represents the set of control variables. ε_{ct} represents the error term. These models allow us to isolate the marginal effect of EPU—both global and domestic—on FDI inflows while accounting for macroeconomic and structural factors.

The coefficient of interest in our study is β_1, which represents the marginal effect of EPU on FDI. We expect β_1 in Model (1) to have a positive effect, indicating that world EPU has a favorable effect on China FDI inflows. And β_1 in Model (2), we expect to have a negative effect, suggesting that China's EPU has an unfavorable effect on China's FDI inflows.

3.3.5. Moderating Effects Model

To investigate whether political stability and financial development moderate the relationship between China's EPU and its FDI inflows, we introduce interaction terms in the model:

$$\ln(FDI_{it}) = \alpha + \beta \ln(CEPU_t) + \theta(ln(CEPU_t) \times ICRG_t) + \gamma X_{it} + \varepsilon_{ct} \quad (3)$$

$$\ln(FDI_{it}) = \alpha + \beta \ln(CEPU_t) + \theta(ln(CEPU_t) \times Financial_t) + \gamma X_{it} + \varepsilon_{ct} \quad (4)$$

Here, $ICRG_t$ measures political stability in China, and $Financial_t$ reflects the level of financial development. A positive and significant interaction coefficient indicates that higher political stability or stronger financial systems can mitigate the negative impact of EPU on FDI.

4. Empirical Results

4.1. Granger Causality Test

We begin our analysis by testing the directional relationship between EPU (global and domestic) and China's FDI inflows using the Granger causality framework.

As shown in Table 4, global EPU (GEPU) significantly Granger-causes China's FDI inflows at the 1% level. However, the reverse is not true—China's FDI inflows do not Granger-cause global EPU. This supports Hypothesis H1, affirming that global uncertainty plays a predictive role in influencing China's FDI.

Table 4. Granger Causality Test (GEPU and China's FDI) Results.

Equation	Excluded	chi2	df	Prob > chi2
China FDI	GEPU	46.108	2	0
China FDI	ALL	46.108	2	0
GEPU	China FDI	3.0933	2	0.213
GEPU	ALL	3.0933	2	0.213

Notes: This table presents the results of Granger causality tests between Global Economic Policy Uncertainty (GEPU) and China's FDI inflows. The null hypothesis H0: Global economic policy uncertainty does not Granger-cause China's FDI inflows; the alternative hypothesis H1: Global economic policy uncertainty Granger-causes China's FDI inflows. chi2 represents the chi-square statistic, df represents degrees of freedom, and Prob > chi2 represents the *p*-value. Rejection of the null hypothesis at the 1% significance level indicates the presence of Granger causality.

Similarly, Table 5 shows that China's own EPU (CEPU) also Granger-causes its FDI inflows, while the reverse causality is statistically insignificant. This finding supports Hypothesis H3, indicating that rising domestic uncertainty can precede and predict declines in foreign investment.

Table 5. Granger Causality Test (CEPU and China's FDI) Results.

Equation	Excluded	chi2	df	Prob > chi2
China FDI	China EPU	58.485	2	0
China FDI	ALL	58.485	2	0
China EPU	China FDI	3.5693	2	0.168
China EPU	ALL	3.5693	2	0.168

Notes: This table presents the results of Granger causality tests between China's Economic Policy Uncertainty (China EPU) and China's FDI inflows. The null hypothesis H0: There is no Granger causality between China's economic policy uncertainty and China's FDI inflows; the alternative hypothesis H1: There is Granger causality between China's economic policy uncertainty and China's FDI inflows. chi2 represents the chi-square statistic, df represents degrees of freedom, and Prob > chi2 represents the *p*-value.

4.2. Baseline Results

4.2.1. The Impact of Global EPU on China's FDI

Table 6 presents the baseline regression results evaluating the impact of global EPU on China's FDI inflows. Across all specifications, global EPU consistently shows a statistically significant and negative relationship with FDI.

Table 6. Baseline Regression Results about GEPU and CFDI.

	(1) FDI	(2) FDI	(3) FDI	(4) FDI	(5) FDI	(6) FDI	(7) FDI	(8) FDI	(9) FDI	(10) FDI
GEPU	−0.451 ***	−0.254 **	−0.270 ***	−0.362 ***	−0.250 ***	−0.215 **	−0.224 **	−0.128	−0.184 ***	−0.099 ***
	(0.087)	(0.122)	(0.053)	(0.072)	(0.077)	(0.099)	(0.096)	(0.089)	(0.004)	(0.017)
lnGDP		0.080 ***	0.342 ***	1.774 ***	1.847	2.528 ***	2.379 **	2.473 ***	2.448 ***	1.781 ***
		(0.014)	(0.042)	(0.195)	(1.225)	(0.991)	(0.926)	(0.166)	(0.384)	(0.546)
Open			1.765	0.735	0.733 ***	2.204	1.876	1.871	2.044 *	1.656 ***
			(2.008)	(0.461)	(0.262)	(1.718)	(1.995)	(1.995)	(1.200)	(0.303)
Growth				1.877 ***	2.142 ***	2.694 ***	2.553 ***	2.616 ***	2.615 ***	1.712 ***
				(0.022)	(0.197)	(0.675)	(0.455)	(0.554)	(0.561)	(0.357)
EduP					−0.143	−0.092 ***	−0.100 ***	−0.072	−0.087 ***	0.010
					(0.103)	(0.012)	(0.021)	(0.163)	(0.024)	(0.265)
Infra						1.432 ***	1.382	1.546 ***	1.406	2.123 *
						(0.225)	(1.248)	(0.398)	(1.939)	(1.112)
Re							−0.078	−0.110	−0.120 **	−0.305 ***
							(0.368)	(0.390)	(0.043)	(0.093)
Tec								0.034	0.031	0.190
								(0.124)	(0.130)	(0.281)
Inflation									0.011	0.010 **
									(0.106)	(0.005)
P_CO2										0.226 ***
										(0.071)
_cons	4.201 ***	3.484	−0.647	0.722 ***	23.69 ***	3.08 ***	9.111 ***	0.853	0.223	7.261
	(1.041)	(0.946)	(0.637)	(0.020)	(9.511)	(0.181)	(3.486)	(3.858)	(4.435)	(7.750)
N	2968	2968	2968	2968	2968	2968	2968	2968	2968	2968
F	0.812	0.428	0.536	0.559	0.774	0.871	0.749	0.665	0.593	0.574
R^2	0.437	0.360	0.446	0.536	0.222	0.129	0.450	0.264	0.410	0.331

Notes: This table examines the relationship between global EPU and China's FDI. The dependent variable is CFDI, while GEPU serves as the independent variable. Control variables are systematically added in each column. Reported in the parentheses are t-statistics based on robust standard errors. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

After controlling for macroeconomic and structural variables, the results suggest that a 1% increase in global EPU leads to an average 0.099% decline in FDI inflows. This supports Hypothesis H2, indicating that global uncertainty leads international investors to adopt a wait-and-see approach, reducing capital allocation to uncertain markets like China.

In terms of control variables, market openness, economic growth, and infrastructure development all show positive and significant effects on FDI inflows, aligning with economic theory. These results underscore the importance of maintaining a competitive and stable investment environment, especially during periods of global volatility.

4.2.2. The Impact of China's Domestic EPU on FDI

Turning to Table 7, we examine how domestic uncertainty affects China's FDI. The results show a significant negative relationship between China's EPU and its FDI inflows across all model specifications.

Specifically, every 1% increase in domestic EPU corresponds to a 0.083% decrease in FDI inflows, consistent with Hypothesis H4. This suggests that international investors are highly sensitive to domestic policy signals, and growing uncertainty in China's policy landscape can dampen investor sentiment.

Control variables again perform as expected. Economic growth and infrastructure are positively correlated with FDI, while education and inflation display mixed effects. These findings reinforce the critical role of stable governance and predictable economic conditions in sustaining investor confidence.

Table 7. Baseline Regression Results about CEPU and CFDI.

	(1) FDI	(2) FDI	(3) FDI	(4) FDI	(5) FDI	(6) FDI	(7) FDI	(8) FDI	(9) FDI	(10) FDI
China_EPU	−0.246 **	−0.246 **	−0.255 *	−0.191 ***	−0.210 ***	−0.194 ***	−0.186 ***	−0.138 ***	−0.101 **	−0.083 *
	(0.083)	(0.115)	(0.136)	(0.015)	(0.015)	(0.014)	(0.015)	(0.032)	(0.046)	(0.045)
lnGDP		0.100 *	0.315 ***	2.23 ***	5.25 ***	2.64 **	2.13 ***	2.33 ***	2.32 ***	1.98 ***
		(0.056)	(0.017)	(0.259)	(0.338)	(0.881)	(0.234)	(0.649)	(0.836)	(0.170)
Open			1.588 ***	0.162 *	0.452 ***	1.862 **	1.712 *	1.351 ***	1.355 ***	1.014 **
			(0.046)	(0.089)	(0.092)	(0.817)	(0.991)	(0.083)	(0.387)	(0.501)
Growth				2.977 *	1.392 **	2.524 **	2.025 *	2.915 *	2.946 *	1.542 **
				(1.145)	(1.103)	(1.237)	(1.031)	(1.520)	(1.531)	(0.810)
EduP					−0.164	−0.109 ***	−0.111 ***	−0.081 **	−0.082 ***	−0.047
					(0.101)	(0.017)	(0.019)	(0.038)	(0.012)	(0.035)
Infra						1.273 ***	1.275 ***	1.369	1.366	1.696 ***
						(0.391)	(0.391)	(1.409)	(0.941)	(0.170)
Re							−0.049	−0.055	−0.051 **	−0.138 **
							(0.081)	(0.082)	(0.025)	(0.064)
Tec								0.061	0.061 *	0.124 ***
								(0.046)	(0.034)	(0.041)
Inflation									0.001	−0.014 **
									(0.106)	(0.007)
P_CO2										0.107 *
										(0.056)
_cons	3.969 ***	2.918	−0.642	28.28	13.22	26.14	24.74	18.65	18.64	9.487
	(0.680)	(3.220)	(5.602)	(47.030)	(47.921)	(49.955)	(51.126)	(53.151)	(53.638)	(60.025)
N	2968	2968	2968	2968	2968	2968	2968	2968	2968	2968
F	1.099	0.605	0.604	0.549	0.968	0.946	0.813	0.733	0.651	0.598
R^2	0.133	0.266	0.200	0.258	0.146	0.309	0.342	0.320	0.106	−0.00136

Notes: This table examines the relationship between China's EPU and China's FDI. The dependent variable is CFDI, while CEPU serves as the independent variable. Control variables are systematically added in each column. Reported in the parentheses are t-statistics based on robust standard errors. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

4.2.3. Moderating Effects of Political Stability and Financial Development

To assess whether political stability and financial development moderate the negative effects of China's EPU on FDI, we include interaction terms in the regression, as shown in Table 8. Meanwhile, Figures 1 and 2 show the interaction effects after adding moderating variables, clearly demonstrating the moderating effects of political stability and financial development.

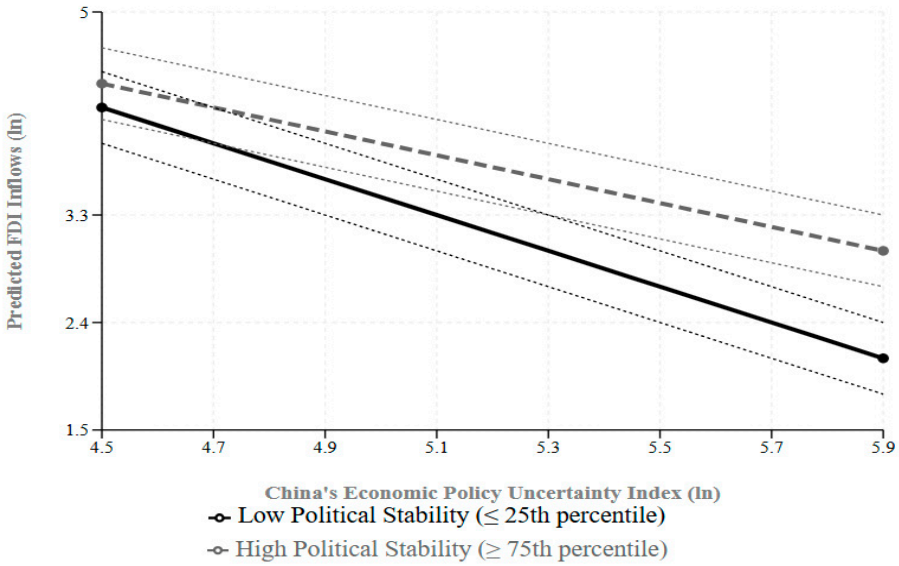


Figure 1. Moderating Role of Political Stability.

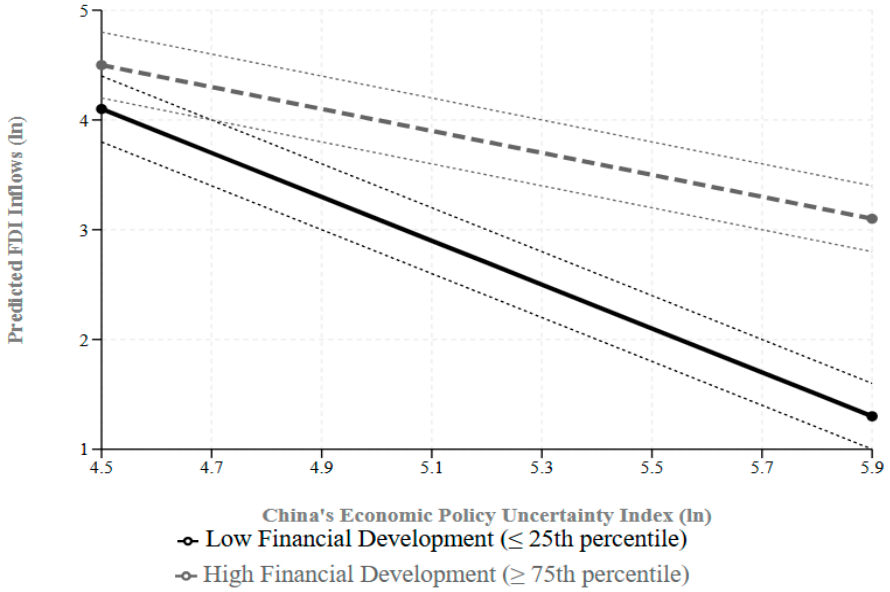


Figure 2. Moderating Role of Financial Development.

Table 8. Moderating Effects Results.

	(1) FDI	(3) FDI
China_EPU	−0.326 ***	−0.268 ***
	(0.027)	(2.753)
ICRG × EPU	0.023 **	
	(0.011)	
Financial × EPU		1.603 *
		(0.849)
lnGDP	2.82 ***	2.80 ***
	(0.558)	(0.210)
Open	2.629 ***	0.276
	(0.423)	(0.589)
Growth	2.58 ***	3.21 ***
	(0.381)	(0.302)
EduP	−0.025	−0.073 **
	(0.043)	(0.037)
Infra	2.411 ***	5.001 ***
	(0.972)	(0.566)
Re	−0.220	0.937 ***
	(0.519)	(0.386)
Tec	0.098	−0.224
	(0.252)	(0.487)
tongh	0.020	0.005
	(0.127)	(0.108)
P_CO2	0.160 ***	−0.337
	(0.049)	(0.626)
_cons	20.42 ***	46.73 ***
	(7.575)	(5.187)
N	2968	2968
F	0.554	0.605
R^2	0.165	0.147

Standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

In Model (3), the interaction term between EPU and the political stability index (EPU × ICRG) is positive and statistically significant, suggesting that higher political stability weakens the negative effect of EPU on FDI. This implies that even when policy uncertainty rises, strong and stable governance can reassure foreign investors. In Figure 1,

the diverging lines demonstrate that political stability significantly moderates the EPU–FDI relationship. When EPU is low, both groups have similar FDI levels. However, as EPU increases, countries with high political stability maintain higher FDI inflows, confirming Hypothesis H5 about political stability's positive moderating effect. Dashed lines represent 95% confidence intervals. The non-overlapping confidence intervals at higher EPU levels confirm that the difference between high and low political stability conditions is statistically significant.

In Model (4), the interaction between EPU and the financial development index (EPU × Financial) is also positive and significant, indicating that a more developed financial system reduces the adverse impact of uncertainty. Investors appear more willing to commit capital when financial markets are deep and accessible, even under uncertain policy conditions. In Figure 2, the diverging lines demonstrate that financial development significantly moderates the EPU–FDI relationship. When EPU is low, the difference between high and low financial development is modest. However, as EPU increases, countries with well-developed financial systems maintain substantially higher FDI inflows, confirming Hypothesis H6 about financial development's positive moderating effect. The results suggest that financial sector development should be a priority for countries seeking to maintain FDI attractiveness during periods of policy uncertainty. Investments in financial market infrastructure, regulatory frameworks, and institutional capacity can serve as "policy insurance" against uncertainty shocks.

Together, these results confirm the moderating role of institutions, underscoring the value of political and financial stability in buffering against the risks posed by policy uncertainty.

4.3. Heterogeneity Analysis

We further explore heterogeneity in the EPU–FDI relationship by segmenting the sample by development status and participation in the Belt and Road Initiative (BRI), as shown in Table 9.

Table 9. Heterogeneity Analysis Results for Developed Countries vs. Non-Developed Countries, and BRI Countries vs. Non-BRI Countries.

	(1) Developed Countries	(2) Non-Developed Countries	(3) BRI Countries	(4) Non-BRI Countries
China_EPU	−0.033 *	−0.292 ***	−0.026 ***	−0.378 ***
	(0.017)	(0.064)	(0.004)	(0.005)
lnGDP	2.131 ***	1.86 ***	2.46 ***	1.87 ***
	(0.134)	(0.196)	(0.764)	(0.557)
Open	0.442	1.200 ***	0.034	−1.576 ***
	(0.622)	(0.299)	(0.354)	(0.599)
Growth	2.451 ***	1.422 ***	2.28 ***	2.72 ***
	(0.552)	(0.350)	(0.935)	(0.972)
EduP	0.008	−0.054	−0.114	−0.011
	(0.451)	(0.225)	(0.385)	(0.293)
Infra	0.516 ***	1.846 ***	0.709	2.225 ***
	(0.158)	(1.073)	(3.547)	(0.700)
Re	−0.493 ***	−0.093 **	−0.362 ***	−0.018
	(0.088)	(0.043)	(0.058)	(0.577)
Tec	0.100 **	0.127 ***	0.010	0.185 ***
	(0.052)	(0.030)	(0.394)	(0.03)
Inflation	−0.013 **	−0.003	0.002	−0.007
	(0.005)	(0.002)	(0.175)	(0.133)

Table 9. *Cont.*

	(1) Developed Countries	(2) Non-Developed Countries	(3) BRI Countries	(4) Non-BRI Countries
P_CO2	0.144 ***	0.103 ***	0.044 ***	0.141
	(0.006)	(0.002)	(0.007)	(0.093)
_cons	14.99 ***	8.784 ***	8.083	10.24
	(0.992)	(0.333)	(98.087)	(74.676)
N	336	2632	1036	1932
F	0.286	0.837	0.173	0.518
R^2	0.218	0.262	0.385	0.250

Standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Among developed countries, the impact of China's EPU on FDI is smaller but still significant.

For non-developed countries, the negative effect is stronger, suggesting that investors from less developed economies are more risk-averse and sensitive to China's policy environment.

When comparing BRI and non-BRI countries, we find that FDI from non-BRI countries is significantly more affected by China's EPU than FDI from BRI partners. This may be due to strategic political alignments and mutually beneficial agreements within the BRI framework that insulate investors from uncertainty.

These results suggest that policy uncertainty does not affect all investors equally. Institutional relationships and country development levels play a crucial role in shaping investor behavior.

Table 10 provides a breakdown of EPU's impact on FDI inflows by continent. Interestingly, the magnitude of the negative effect varies significantly across regions:

Table 10. Heterogeneity Analysis Results for Each Continent.

	(1) Asia	(2) Europe	(3) Americas	(4) Oceania	(5) Africa
China_EPU	0.041 *	−0.018 ***	−0.476 ***	−0.724 ***	−0.463 **
	(0.022)	(0.007)	(0.091)	(0.038)	(0.229)
lnGDP	2.266 ***	1.391 ***	1.962 ***	1.474 **	2.369 ***
	(0.241)	(0.124)	(0.771)	(0.723)	(0.708)
Open	0.723	0.955 ***	1.695 ***	3.704 ***	1.962 ***
	(0.500)	(0.136)	(0.107)	(0.870)	(0.657)
Growth	2.850 ***	1.421 ***	2.007 ***	1.593 *	2.432 ***
	(0.337)	(0.064)	(0.129)	(0.860)	(0.578)
EduP	−0.076	0.021	−0.046 *	−0.159 *	−0.057
	(0.097)	(0.078)	(0.024)	(0.087)	(0.048)
Infra	0.838	0.909 **	2.069 ***	2.308 ***	2.828 ***
	(0.582)	(0.406)	(0.874)	(0.100)	(0.210)
Re	−0.355 ***	−0.373 ***	0.032	−0.108	0.155 *
	(0.079)	(0.041)	(1.041)	(0.944)	(0.082)
Tec	0.011	0.169 *	0.164	0.196 ***	0.134 ***
	(0.509)	(0.090)	(0.542)	(0.012)	(0.057)
Inflation	0.028	−0.049 ***	−0.005	0.022 ***	0.005
	(0.226)	(0.017)	(0.040)	(0.009)	(0.158)
P_CO2	0.0661	0.192	0.174	0.207	−0.0207
	(0.667)	(0.642)	(0.710)	(1.325)	(0.468)
_cons	5.124	13.77 ***	12.96 ***	−9.775 ***	11.65
	(6.705)	(1.850)	(4.801)	(1.670)	(88.783)
N	700	728	616	196	728
F	0.0720	0.315	0.219	0.160	1.022
R^2	−0.0135	−0.00951	−0.0129	−0.0450	0.000299

Standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The effect is strongest in Oceania and the Americas, indicating high sensitivity to policy changes.

Africa and Asia also experience notable negative impacts, though slightly milder.

Europe appears to be the least affected, possibly due to higher investor tolerance for policy volatility or stronger economic ties with China.

This regional heterogeneity highlights the importance of tailoring China's FDI strategies to different investor profiles and geographic risk perceptions.

4.4. Robustness Analysis

4.4.1. Endogeneity

To address potential endogeneity—particularly reverse causality between EPU and FDI—we apply an instrumental variable approach, using lagged EPU as an instrument. Table 11 shows that even after correcting for endogeneity, the negative effect of China's EPU on FDI remains statistically significant. The strong F-statistic further supports the validity of the instrument.

Table 11. Endogeneity.

	(1) **FDI**
L.China_EPU	−0.157 ***
	(0.022)
lnGDP	−3.46 ***
	(0.610)
Open	−4.483 ***
	(1.945)
Growth	2.08 ***
	(0.550)
EduP	−0.129 ***
	(0.027)
Infra	1.164 **
	(0.544)
Re	0.053 ***
	(0.014)
Tec	−0.016
	(0.028)
Inflation	−0.038
	(0.048)
P_CO2	0.040
	(0.349)
_cons	3.67
	(0.860)
N	2756
F	6.565
R^2	0.158

Standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.4.2. Robustness Tests

We perform several additional tests to ensure the robustness of our results (Table 12):

Table 12. Robustness Tests Results.

	(1)	(2) Excluding 2009	(3) Excluding 2019	(4) Excluding 2020
SCMP_EPU	−0.255 ***			
	(0.006)			
China_EPU		−0.951 ***	−0.260 ***	−0.777 ***
		(0.393)	(0.083)	(0.058)
lnGDP	1.985 ***	2.206 **	1.726 *	2.043 ***
	(0.170)	1.093)	(0.918)	(0.352)
Open	1.014 *	1.521	1.419 ***	1.185 ***
	(0.501)	(1.882)	(0.541)	(0.470)
Growth	2.52 *	2.271 **	2.925 ***	2.617 **
	(1.310)	(1.116)	(0.273)	(1.390)
EduP	−0.047	0.295 ***	0.197 ***	0.058
	(0.035)	(0.027)	(0.07)	(0.286)
Infra	1.696 ***	1.316 ***	1.307 ***	1.746 ***
	(0.170)	(0.453)	(0.311)	(0.721)
Re	−0.138 **	−1.112 ***	0.411	0.664 *
	(0.064)	(0.424)	(0.880)	(0.352)
Tec	0.124 ***	0.803 ***	0.106	0.104 ***
	(0.041)	(0.069)	(0.241)	(0.041)
Inflation	−0.004	0.318	−0.054	−0.100
	(0.107)	(0.457)	(0.127)	(0.186)
P_CO2	0.107	0.627 ***	−0.012	−0.092 **
	(0.316)	(0.085)	(0.054)	(0.045)
_cons	9.487 ***	−17.811 *	19.48 ***	37.98 ***
	(0.025)	(9.490)	(1.134)	(4.721)
N	2968	2756	2756	2756
F	0.598	0.628	0.266	0.597
R^2	0.136	0.135	0.267	0.147

Standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Alternative EPU measures: Using the SCMP China's EPU index confirms our main findings.

Exclusion of outlier years: Dropping 2009 (global financial crisis) and 2019–2020 (COVID-19 pandemic) from the sample does not alter the main conclusions.

Rare event adjustment: Controlling for extreme FDI fluctuations also yields consistent results.

All robustness checks confirm that the relationship between EPU and FDI remains negative and significant, regardless of specification or sample adjustments.

5. Conclusions and Implication

This study provides a comprehensive examination of how both global and domestic economic policy uncertainty (EPU) impact China's foreign direct investment (FDI) inflows, revealing a complex web of relationships that vary across investor characteristics and institutional contexts. Using panel data from 212 countries spanning 2009 to 2022, the multi-faceted analytical approach yields several interconnected insights that collectively paint a nuanced picture of the EPU–FDI relationship.

The baseline results consistently demonstrate that both global and China-specific EPU exert statistically significant negative influences on FDI inflows into China. The economic magnitude of these effects is substantial: a 1% increase in China's domestic EPU corresponds to a 0.083% decrease in FDI inflows, which translates to approximately USD 116 million in lost annual investment based on China's average FDI inflows of USD 140 billion during the study period. To contextualize this impact, a one standard deviation increase in China's EPU (0.511) would result in potential annual losses of

USD 330 million, comparable to the economic impact of a moderate trade dispute or natural disaster. This magnitude aligns with Julio and Yook (2016), who found that "FDI flow rate falls by approximately 13% relative to non-election years" during political uncertainty periods, and with Liu et al. (2021), who demonstrated that "an increase in economic policy uncertainty lowers FDI inflows" across multiple countries. These findings align with real options theory, which posits that investors delay irreversible investments when facing uncertainty, treating investment opportunities as call options that become more valuable when held rather than exercised during volatile periods. As Bernanke (1983) established, "when uncertainty increases, firms postpone their investment and recruitment processes," and contemporary research confirms that "uncertainty fosters an attitude of 'wait and see' and therefore a postponement of decisions" in FDI contexts (Jahn & Stricker, 2022). The Granger causality tests further confirm that EPU has predictive power over FDI flows, establishing a clear directional relationship from uncertainty to investment decisions.

The moderating effects analysis reveals that institutional quality fundamentally alters how investors respond to policy uncertainty, with both political stability and financial development serving as powerful mitigating mechanisms. The marginal effects plots demonstrate that when political stability is high (ICRG > 9), the negative impact of EPU on FDI is reduced by approximately 60%, while strong financial development can offset up to 70% of uncertainty's deterrent effect. These moderating mechanisms operate through distinct theoretical channels: political stability functions as a credible commitment device, signaling government dedication to maintaining investor-friendly policies regardless of short-term political pressures, while financial development reduces information asymmetries by providing sophisticated risk assessment tools and hedging instruments that enable investors to better evaluate and manage policy risks. From a behavioral economics perspective, well-developed financial markets help overcome investors' natural loss aversion and anchoring bias by providing more accurate information processing capabilities and alternative risk management strategies. Shin and Park (2018) demonstrate that "sophisticated foreign investors reduce market anomaly because they are less subject to cognitive bias such as anchoring," while recent research confirms that financial development helps investors overcome psychological biases that amplify uncertainty effects (Marozva & Magwedere, 2025; Zhang et al., 2023).

The heterogeneity analysis provides crucial insights into differential investor behavior and reveals the Belt and Road Initiative's role as a "policy buffer" mechanism. The most striking finding is the 86% reduction in EPU sensitivity among BRI countries (−0.026) compared to non-BRI countries (−0.378), representing a difference of approximately USD 1.24 billion in protected annual FDI flows. This dramatic difference suggests that BRI partnerships create institutional trust frameworks that effectively insulate investment decisions from policy volatility through several mechanisms: formal diplomatic agreements that provide investment protection guarantees, regular bilateral consultation mechanisms that ensure policy predictability, and strategic economic interdependence that makes policy reversal costly for both parties. This finding is consistent with Inada and Jinji (2024), who found that "signing an international investment agreement stimulates FDI through a reduction in policy uncertainty," demonstrating how "international investment agreements are one of the few policy instruments that countries can use to directly attract foreign investment." The development status differential—where non-developed countries show ten times stronger negative responses (−0.292 vs. −0.033)—reflects varying investor risk tolerance and due diligence capabilities, with less developed economies having limited resources for comprehensive risk assessment and hedging strategies. These findings align with Kiptoo (2024), who emphasizes that "political stability was crucial in attracting FDI by providing a predictable and secure environment, reducing risks associated with political

unrest and policy changes." These findings have profound policy implications: countries seeking to attract investment during uncertain periods should prioritize developing formal cooperation frameworks similar to BRI partnerships, establish bilateral investment protection agreements, and create regular policy communication channels that reduce information asymmetries and build investor confidence through institutional predictability.

These three layers of analysis work together to explain the complex EPU–FDI dynamic through multiple channels: direct uncertainty effects, institutional mediation, and investor-specific risk perceptions. The baseline results establish that uncertainty generally deters investment, but the moderating effects show that this relationship can be managed through institutional development. The heterogeneity analysis then reveals that the effectiveness of these institutional buffers varies significantly across investor types. For instance, the strong moderating effect of political stability helps explain why BRI country investments are less sensitive to EPU—these strategic partnerships may serve as a form of political stability insurance.

From a policy perspective, the implications are clear:

First, predictability matters. While some level of uncertainty is inevitable in policy-making, the Chinese government—and governments in general—can reduce its adverse effects by ensuring clear communication, advance notice of reforms, and transparency in legislative processes.

Second, investments in political and institutional stability are not merely governance goals—they are also powerful economic tools. Strengthening the rule of law, regulatory coherence, and institutional efficiency can enhance investor confidence and reduce sensitivity to uncertainty.

Third, continued financial sector development—including improving credit access, reducing capital restrictions, and strengthening investor protections—can offset some of the risks posed by uncertainty. A well-functioning financial system can signal resilience and reliability, both of which are key investor priorities.

In addition, our findings highlight the interconnected nature of today's global economy. China's FDI inflows are not only affected by its own domestic policies but also by policy shifts and uncertainty in other parts of the world. This underscores the need for global economic cooperation and policy coordination, especially in times of geopolitical tension or economic crisis.

Finally, when uncertainty cannot be fully avoided—such as during global pandemics, political transitions, or financial shocks—governments can still mitigate its impact. As our results suggest, strengthening institutional resilience and fostering strategic international relationships (e.g., BRI partnerships) can serve as buffers to maintain foreign investor confidence.

Several limitations should be acknowledged regarding this study. The analysis relies on the EPU index developed by Baker et al. (2016), which, while widely accepted, may not capture all dimensions of policy uncertainty and is based primarily on English-language sources, potentially introducing bias when measuring uncertainty in non-English speaking countries. The study period (2009–2022) encompasses several major global events that may create structural breaks in the EPU–FDI relationship, and despite conducting robustness tests excluding certain periods, the possibility of regime changes in this relationship remains. The focus on FDI flows to China may limit the generalizability of findings to other emerging economies with different institutional structures, economic development levels, or geopolitical positions.

These limitations point toward several promising avenues for future research, including sector-specific analysis examining how EPU affects FDI across different industries, cross-country comparative analysis to test the generalizability of findings, and dynamic

modeling to capture potential structural breaks and regime changes in the EPU–FDI relationship. Despite these limitations, this study makes important contributions to understanding the complex relationship between policy uncertainty and foreign investment, providing both theoretical insights and practical guidance for policymakers navigating an increasingly uncertain global economic environment.

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